Pointer

A pointer is a variable that stores the memory address of another variable.

<u>Example program</u>

```
#include<stdio.h>

void main( )

{

        int a = 10 ;

        int *p;   //p is a pointer

        p=&a;    //p holds the address of a, '&' is 'address of' operator

        int **q;   //p is a pointer to a pointer

        q=&p;     //q holds the address of p which is a pointer itself

        printf ( "\nValue of a = %d", a);

        printf ( "\nAddress of a = %u", &a);

        printf ( "\nAddress of a = %u", p);

        printf ( "\nValue of a = %d", *p) ;   //* is called 'value at address' operator.It is also called 'indirection' operator

        printf ( "\nValue of a = %d", *(&a)) ;

        printf ( "\nAddress of p = %u", &p);

        printf ( "\nAddress of p = %u", q);

        printf ( "\nAddress of a = %u", *q);

        printf ( "\nValue of a = %d", **q);

}
```

Sample output of the above program :

```
Value of a = 10
Address of a = 65524
Address of a = 65524
Value of a = 10
Value of a = 10
Address of p = 65522
Address of p = 65522
Address of a = 65524
Value of a = 10
```

C program to swap 2 integers using pointers

```
#include<stdio.h>

void main( )

{

    int a,b,t,*ptr1,*ptr2;

    ptr1=&a;

    ptr2=&b;

    printf("\nEnter the 2 numbers:");

    scanf("%d%d",&a,&b);

    printf("before swapping, a=%d, b=%d",a,b);

    t=*ptr1;

    *ptr1=*ptr2;

    *ptr2=t;

    printf("after swapping, a=%d, b=%d",a,b);

}
```

Pointer arithmetics in C

It is the set of valid arithmetic operations that can be performed on pointers.

1. Addition of an integer to a pointer
2. Subtraction of an integer from a pointer
3. Subtraction of one pointer from another (of the same type)
4. Comparison of two pointer variables

Method of accessing array elements by using subscripted variables

```
int main( )
{
    int arr[ ] = { 10, 5, 42, 54, 96, 15 } ;
    int i ;
    for ( i = 0 ; i <= 5 ; i++ )
    {
        printf ( "\naddress = %u ", &arr[i] ) ;
        printf ( "element = %d", arr[i] ) ;
    }
    return 0;
}
```

Sample Output

```
address = 65522 element = 10
address = 65524 element = 5
address = 65526 element = 42
address = 65528 element = 54
address = 65530 element = 96
address = 65532 element = 15
```

Pointers and Arrays

Method of accessing array elements by using pointers

```
int main( )
{
        int arr[ ] = { 10, 5, 42, 54, 96, 15 } ;
        int i ;
        int *p;
        p=&arr[0];     // assign address of zeroth element of the array
        for ( i = 0 ; i <= 5 ; i++ )
        {
                printf ( "\naddress = %u ", p ) ;
                printf ( "element = %d", *p ) ;
                p++;
        }
        return 0;
}
```

Sample Output

address = 65522 element = 10

address = 65524 element = 5

address = 65526 element = 42

address = 65528 element = 54

address = 65530 element = 96

address = 65532 element = 15

Passing an Array to a Function

```
int main( )
{
        int num[ ] = { 10, 5, 42, 54, 96, 15 } ;
        display ( &num[0], 6 ) ;    //passing the base address and number of elements of the array as parameters
        return 0;
}
void display ( int *p, int n )
{
        int i ;
        for ( i = 0 ; i <= n - 1 ; i++ )
        {
                printf ( "\nelement = %d\n ", *p) ;
                p++ ;           // increment pointer to point to next element
        }
}
```

Output

```
10
5
42
54
96
15
```

Structure pointers (pointer to a structure)

```
int main( )
{

    struct emp
    {
        int age;
        float salary;
        char name[20];
     };

    struct emp e1,e2;
    struct emp *p1,*p2;
    p1=&e1;
    p2=&e2;        //p1 and p2 are pointers to structure

    printf("Enter the age of employee1:");
    scanf("%d",p1->age);
    printf("\nEnter the salary of employee1:");
    scanf("%f",& p1->salary);
    printf("\nEnter the name of employee1:");
    scanf("%s",& p1->name);

    printf("Enter the age of employee2:");
    scanf("%d",& p2->age);
    printf("\nEnter the salary of employee2:");
    scanf("%f",& p2->salary);
    printf("\nEnter the name of employee2:");
    scanf("%s",& p2->name);

    printf("\nDetails of employee1 are:");
    printf("\nAge of employee1 is %d", p1->age);
    printf("\nSalary of employee1 is %f", p1->salary);
    printf("\nName of employee1 is %s", p1->name);

    printf("\nDetails of employee2 are:");
    printf("\nAge of employee2 is %d", p2->age);
    printf("\nSalary of employee2 is %f", p2->salary);
    printf("\nName of employee2 is %s", p2->name);

    return 0;
}
```